

Computer Networking

BCA — Semester V — 2022 — Answer Sheet

Subject Code: CACS303

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

2. What is digital signature and how it works?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Why do we need DNS when we can directly use an IP address? Explain the process to resolve the domain www.tribhuvan-university.edu.np?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. What do you mean by IP datagram fragmentation?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Explain the mechanism of bit and byte stuffing with suitable example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Explain the causes of transmission impairment in data communication.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. What is flow control? Explain Go-Back-N ARQ with suitable diagram.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Explain in detail about the structure of IPv6 address. Differentiate between IPv4 and IPv6 address.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Group C

Attempt any TWO questions[2x10=20]

9. Explain how slotted Aloha improves the performance of system over pure Aloha. A bit stream of 100101010 is transmitted using CRC method. The generator polynomial is $x^3 + 1$. Show the actual bit string transmitted and if fourth bit from LSB is altered, show the error detection in receiver side.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Why do we need layered protocol architecture? Discuss each layer of TCP/IP architecture along with function of each layer. Write the protocols used in each layer of TCP/IP.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Why do we need subnetting? What is the new subnet mask? Write subnet ID and broadcast address of each subnet if you divide a class C network (192.6.3.0–192.6.3.255) into 4 different subnets?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.